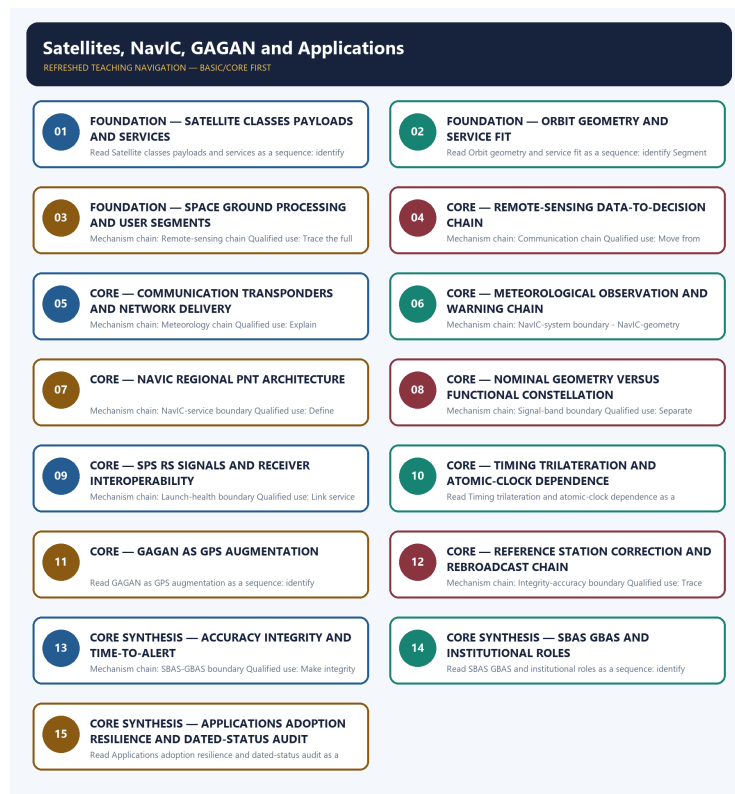


SOLVED PRACTICE WORKBOOK

Satellites, NavIC, GAGAN and Applications - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Satellite-class boundary?

A. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. B. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. C. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. D. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.

Answer: A. Explanation: Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Satellite-class boundary?

A. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. B. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. C. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. D. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.

Answer: B. Explanation: Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Satellite-class boundary without changing its institution, unit or status?

A. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. B. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. C. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. D. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.

Answer: C. Explanation: Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Satellite-class boundary?

A. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. B. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. C. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: D. Explanation: Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. The other options change the

scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Orbit-service boundary?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. D. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.

Answer: A. Explanation: A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Orbit-service boundary?

A. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. B. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. C. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. D. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.

Answer: B. Explanation: A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Orbit-service boundary without changing its institution, unit or status?

A. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. B. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. C. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. D. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.

Answer: C. Explanation: A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Orbit-service boundary?

A. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. B. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service.

Answer: D. Explanation: A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Segment boundary?

A. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. B. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. C. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. D. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.

Answer: A. Explanation: A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Segment boundary?

A. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. D. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Answer: B. Explanation: A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Segment boundary without changing its institution, unit or status?

A. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. B. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. C. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. D. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Answer: C. Explanation: A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Segment boundary?

A. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.

Answer: D. Explanation: A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Remote-sensing chain?

A. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. B. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. C. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. D. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.

Answer: A. Explanation: Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Remote-sensing chain?

A. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. B. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.

Answer: B. Explanation: Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Remote-sensing chain without changing its institution, unit or status?

A. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. D. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health.

Answer: C. Explanation: Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Remote-sensing chain?

A. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.

Answer: D. Explanation: Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Communication chain?

A. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. B. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Answer: A. Explanation: A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Communication chain?

A. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. B. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. C. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. D. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.

Answer: B. Explanation: A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Communication chain without changing its institution, unit or status?

A. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. B. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. C. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. D. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.

Answer: C. Explanation: A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Communication chain?

A. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. B. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. C. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. D. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.

Answer: D. Explanation: A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Meteorology chain?

A. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. B. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.

Answer: A. Explanation: Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Meteorology chain?

A. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. B. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. C. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. D. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.

Answer: B. Explanation: Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Meteorology chain without changing its institution, unit or status?

A. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. D. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.

Answer: C. Explanation: Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Meteorology chain?

A. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. B. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. C. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. D. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.

Answer: D. Explanation: Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies NavIC-system boundary?

A. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.

Answer: A. Explanation: NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of NavIC-system boundary?

A. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. B. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. C. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. D. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.

Answer: B. Explanation: NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses NavIC-system boundary without changing its institution, unit or status?

A. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. B. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. C. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. D. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.

Answer: C. Explanation: NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about NavIC-system boundary?

A. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. B. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. C. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. D. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Answer: D. Explanation: NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies NavIC-geometry boundary?

A. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. B. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. C. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. D. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.

Answer: A. Explanation: The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of NavIC-geometry boundary?

A. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. B. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. C. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. D. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.

Answer: B. Explanation: The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses NavIC-geometry boundary without changing its institution, unit or status?

A. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. B. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. C. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. D. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.

Answer: C. Explanation: The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about NavIC-geometry boundary?

A. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. B. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. C. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. D. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health.

Answer: D. Explanation: The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. The other options change the scientific category, institutional owner, measurement, mission rung or

Q33. Which statement correctly identifies NavIC-service boundary?

A. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. B. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. C. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. D. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.

Answer: A. Explanation: Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of NavIC-service boundary?

A. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. B. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. C. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: B. Explanation: Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses NavIC-service boundary without changing its institution, unit or status?

A. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. B. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. C. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: C. Explanation: Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about NavIC-service boundary?

A. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. B. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. C. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. D. Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.

Answer: D. Explanation: Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. The

other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Signal-band boundary?

A. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. B. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. C. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: A. Explanation: NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Signal-band boundary?

A. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. B. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. C. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. D. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.

Answer: B. Explanation: NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Signal-band boundary without changing its institution, unit or status?

A. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. B. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. C. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. D. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.

Answer: C. Explanation: NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Signal-band boundary?

A. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. D. NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.

Answer: D. Explanation: NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Launch-health boundary?

A. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. B. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. C. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: A. Explanation: Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Launch-health boundary?

A. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. B. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. C. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. D. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.

Answer: B. Explanation: Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Launch-health boundary without changing its institution, unit or status?

A. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. B. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. C. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. D. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.

Answer: C. Explanation: Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Launch-health boundary?

A. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. B. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. C. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. D. Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.

Answer: D. Explanation: Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies PNT-mechanism boundary?

A. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. B. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. C. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: A. Explanation: A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of PNT-mechanism boundary?

A. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. B. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. C. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. D. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation.

Answer: B. Explanation: A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses PNT-mechanism boundary without changing its institution, unit or status?

A. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. B. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. C. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. D. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation.

Answer: C. Explanation: A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about PNT-mechanism boundary?

A. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. D. A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service.

Answer: D. Explanation: A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies GAGAN-system boundary?

A. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. B. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. C. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. D. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation.

Answer: A. Explanation: GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of GAGAN-system boundary?

A. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. B. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. C. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: B. Explanation: GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses GAGAN-system boundary without changing its institution, unit or status?

A. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: C. Explanation: GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about GAGAN-system boundary?

A. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. B. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. C. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. D. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

Answer: D. Explanation: GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies SBAS architecture?

A. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. D. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.

Answer: A. Explanation: Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of SBAS architecture?

A. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. B. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. C. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: B. Explanation: Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses SBAS architecture without changing its institution, unit or status?

A. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. B. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. C. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. D. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions.

Answer: C. Explanation: Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about SBAS architecture?

A. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. B. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. C. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. D. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.

Answer: D. Explanation: Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Integrity-accuracy boundary?

A. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: A. Explanation: Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Integrity-accuracy boundary?

A. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. B. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. C. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: B. Explanation: Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Integrity-accuracy boundary without changing its institution, unit or status?

A. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. B. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. C. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. D. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Answer: C. Explanation: Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Integrity-accuracy boundary?

A. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. D. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.

Answer: D. Explanation: Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies SBAS-GBAS boundary?

A. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. B. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. C. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. D. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Answer: A. Explanation: SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of SBAS-GBAS boundary?

A. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. B. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. C. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. D. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions.

Answer: B. Explanation: SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses SBAS-GBAS boundary without changing its institution, unit or status?

A. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: C. Explanation: SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about SBAS-GBAS boundary?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. D. SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation.

Answer: D. Explanation: SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Institution-role boundary?

A. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. B. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. C. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. D. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence.

Answer: A. Explanation: ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Institution-role boundary?

A. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. B. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. C. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. D. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence.

Answer: B. Explanation: ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Institution-role boundary without changing its institution, unit or status?

A. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. D. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service.

Answer: C. Explanation: ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Institution-role boundary?

A. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. D. ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Answer: D. Explanation: ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Application-outcome boundary?

A. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: A. Explanation: Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Application-outcome boundary?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. C. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. D. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence.

Answer: B. Explanation: Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Application-outcome boundary without changing its institution, unit or status?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: C. Explanation: Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Application-outcome boundary?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. D. Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Answer: D. Explanation: Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Privacy-resilience boundary?

A. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. D. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service.

Answer: A. Explanation: Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Privacy-resilience boundary?

A. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. B. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. C. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: B. Explanation: Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Privacy-resilience boundary without changing its institution, unit or status?

A. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. B. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. C. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. D. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.

Answer: C. Explanation: Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Privacy-resilience boundary?

A. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. D. Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions.

Answer: D. Explanation: Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile constellation boundary?

A. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. B. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. C. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. D. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.

Answer: A. Explanation: Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile constellation boundary?

A. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. B. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. C. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. D. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.

Answer: B. Explanation: Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile constellation boundary without changing its institution, unit or status?

A. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. B. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. C. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. D. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.

Answer: C. Explanation: Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. The other options change the scientific category, institutional owner, measurement, mission rung or

operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile constellation boundary?

A. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. B. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. C. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. D. Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence.

Answer: D. Explanation: Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the 2018 IRNSS Mains demand, 2025 GAGAN objective concept and related navigation, remote-sensing and space-weather objective concepts here. Objective keys are not inferred.

PYQ DEMAND CARD 1 - 2018 GS-I

Demand: Explain why India needs IRNSS and how it helps navigation.

Status: Verified routed Mains demand.

Model solution: NavIC-system boundary: NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **NavIC-geometry boundary:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **NavIC-service boundary:** Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. **PNT-mechanism boundary:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Application-outcome boundary:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Privacy-resilience boundary:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2025 Prelims GS-I

Demand: Identify the nature and function of GAGAN.

Status: Verified routed objective concept; the official Set-A key exists locally but no answer letter is reproduced here.

Model solution: GAGAN-system boundary: GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.

SBAS architecture: Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.

Integrity-accuracy boundary: Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. **SBAS-GBAS boundary:** SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. **Institution-role boundary:** ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator

are distinct roles. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2018-2023 Prelims GS-I

Demand: GPS applications, IRNSS geometry, remote-sensing uses, solar-flare effects and independent navigation systems.

Status: Verified routed concept family; unkeyed rows are not converted into answer letters.

Model solution: **Orbit-service boundary:** A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. **Remote-sensing chain:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **NavIC-system boundary:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **NavIC-geometry boundary:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **PNT-mechanism boundary:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Privacy-resilience boundary:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish NavIC from GAGAN in architecture, purpose and strategic value. Answer in about 150 words.

Model thesis: **Claim:** NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-geometry boundary. **Named evidence/example:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS architecture. **Named evidence/example:** Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Integrity-accuracy boundary. **Named evidence/example:** Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Institution-role boundary. **Named evidence/example:** ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.
- The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health.
- GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.
- Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.
- Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.
- ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.

Qualified conclusion: **Claim:** NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-geometry boundary. **Named evidence/example:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS architecture. **Named evidence/example:** Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Integrity-accuracy boundary. **Named evidence/example:** Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Institution-role boundary. **Named evidence/example:** ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain why a satellite launch is not the same as a satellite service. Answer in about 150 words.

Model thesis: **Claim:** Satellite-class boundary. **Named evidence/example:** Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbit-service boundary. **Named evidence/example:** A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or

experimental-status boundary. **Claim:** Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named evidence/example:** A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.
- A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service.
- A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.
- Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.
- A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.
- Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.

Qualified conclusion: **Claim:** Satellite-class boundary. **Named evidence/example:** Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Orbit-service boundary. **Named evidence/example:** A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named**

evidence/example: A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse the space-ground-user chain behind Indian satellite applications. Answer in about 250 words.

Model thesis: **Claim:** Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named evidence/example:** A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.
- Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.
- A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.
- Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.
- Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Qualified conclusion: **Claim:** Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's

date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named evidence/example:** A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess constellation health and receiver adoption as constraints on NavIC. Answer in about 250 words.

Model thesis: **Claim:** NavIC-geometry boundary. **Named evidence/example:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-service boundary. **Named evidence/example:** Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Signal-band boundary. **Named evidence/example:** NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-health boundary. **Named evidence/example:** Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PNT-mechanism boundary. **Named evidence/example:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health.
- Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.
- NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.
- Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.
- A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service.
- Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

Qualified conclusion: Claim: NavIC-geometry boundary. **Named evidence/example:** The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: NavIC-service boundary. **Named evidence/example:** Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Signal-band boundary. **Named evidence/example:** NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Launch-health boundary. **Named evidence/example:** Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: PNT-mechanism boundary. **Named evidence/example:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate satellite systems as public-service and strategic infrastructure. Answer in about 300 words.

Model thesis: Claim: Satellite-class boundary. **Named evidence/example:** Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim: Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named evidence/example:** A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Privacy-resilience boundary. **Named evidence/example:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name.
- A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service.
- Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions.
- A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals.
- Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.
- NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.
- GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.
- Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome.

- Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions.

Qualified conclusion: Claim: Satellite-class boundary. **Named evidence/example:** Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Segment boundary. **Named evidence/example:** A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Remote-sensing chain. **Named evidence/example:** Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Communication chain. **Named evidence/example:** A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Meteorology chain. **Named evidence/example:** Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Application-outcome boundary. **Named evidence/example:** Navigation, fisheries, transport, timing, weather warning, mapping and disaster support require standards, receivers, data access and departmental adoption; signal or data availability alone is not an outcome. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Privacy-resilience boundary. **Named evidence/example:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a resilient PNT and aviation-augmentation strategy for India. Answer in about 300 words.

Model thesis: Claim: NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

Analysis: This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-service boundary. **Named evidence/example:** Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. **Analysis:** This fixes the scientific

process, system boundary, institution, application and verified capability rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Signal-band boundary. **Named evidence/example:** NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-health boundary. **Named evidence/example:** Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PNT-mechanism boundary. **Named evidence/example:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS architecture. **Named evidence/example:** Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Integrity-accuracy boundary. **Named evidence/example:** Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS-GBAS boundary. **Named evidence/example:** SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Institution-role boundary. **Named evidence/example:** ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Privacy-resilience boundary. **Named evidence/example:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile constellation boundary. **Named evidence/example:** Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation.

- Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military.
- NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption.
- Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source.
- A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service.
- GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC.
- Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast.
- Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.
- SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation.
- ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles.
- Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions.
- Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence.

Qualified conclusion: Claim: NavIC-system boundary. **Named evidence/example:** NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** NavIC-service boundary. **Named evidence/example:** Standard Positioning Service is the open layer, while Restricted Service is encrypted and available to authorised users; open versus restricted is not safely reduced to civilian versus military. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Signal-band boundary. **Named evidence/example:** NavIC's original L5 and S-band signals and the later civil L1 addition are signal and receiver-interoperability choices; a launched L1-capable spacecraft does not prove universal handset adoption. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Launch-health boundary. **Named evidence/example:** Cumulative satellites launched, satellites in intended orbit, satellites operational, satellites broadcasting messages and satellites providing full PNT are different counts that require a dated source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** PNT-mechanism boundary. **Named evidence/example:** A receiver derives position from signal travel time and known satellite positions, so precise timing and atomic-clock health are foundational to navigation service. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** GAGAN-system boundary. **Named evidence/example:** GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS architecture. **Named evidence/example:** Reference stations measure GNSS errors, master control

computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Integrity-accuracy boundary. **Named evidence/example:** Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** SBAS-GBAS boundary. **Named evidence/example:** SBAS provides wide-area satellite-broadcast augmentation, while GBAS provides local airport-area corrections by ground radio; neither is a sovereign navigation constellation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Institution-role boundary. **Named evidence/example:** ISRO and AAI jointly developed GAGAN, AAI owns the aviation operational side, and DGCA certifies aviation performance; developer, operator and regulator are distinct roles. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Privacy-resilience boundary. **Named evidence/example:** Location services can improve logistics and emergency response while also raising surveillance, spoofing, jamming and critical-infrastructure resilience questions. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile constellation boundary. **Named evidence/example:** Constellation composition, satellite health, frequencies, coverage, accuracy, certification, service entry and adoption require dated ISRO, AAI, DGCA or Department of Space evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.